Integrated Rocket Anomaly Diagnostic Report

SpaceX Starship Flight 12 - Super Heavy Booster Flyback Anomaly

Field	Entry
Event date	May 22, 2026
Mission phase	Post-stage-separation booster flyback / return trajectory
Report version	Synthesized Report v1.0
Evidence cutoff	June 15, 2026
Authorization status	Advisory-only / causal findings export-held
Root-cause status	Unresolved

Core rule: This report does not determine root cause. Candidate causal states remain uncommitted absent telemetry, official investigation findings, and authorization support.

Prepared as an audit-ready advisory diagnostic report under the DI Engineering Nonprovisional controlled diagnostic-output framework. Live factual claims are sourced to official FAA and SpaceX public materials, and framework claims are tied to the uploaded DI Engineering filing.

2. Executive Summary

The FAA determined on May 27, 2026 that SpaceX Starship Flight 12 resulted in a mishap involving the Super Heavy booster during flyback over the Gulf of America after stage separation. FAA reported no public injury or public-property damage, required SpaceX to conduct a mishap investigation, and stated that FAA will oversee the investigation and approve SpaceX's final report and corrective actions. FAA also states that return to flight depends on FAA determining that any system, process, or procedure related to the mishap does not affect public safety. [FAA-1]

Verified: FAA mishap determination, Super Heavy booster flyback involvement, no reported public injury or public-property damage, debris/hazard-area response, and FAA-overseen investigation requirement.

Inferred: The propulsion-side candidate is strengthened because SpaceX's official Flight 12 page states the booster was unable to light all planned engines, performed a partial boostback burn that ended early, and attempted to reignite engines for landing burn before hard splashdown. This supports a candidate propulsion sequence issue, but not a committed root cause. [SPX-1]

Unresolved: initiating cause, failed component, engine-health timeline, command-state alignment, GNC response, corrective actions, replay validation, and return-to-flight clearance.

Bottom line: This report supports an audit-ready anomaly review, not a root-cause report.

Reference Sources

ID	Source	Use
FAA-1	Federal Aviation Administration, General Statements, May 22 and May 27, 2026 Flight 12 entries. URL: https://www.faa.gov/newsroom/statements/general-statements	Official status, mishap determination, public-safety outcome, debris response, investigation and return-to-flight gates.
SPX-1	SpaceX, Starship's Twelfth Flight Test. URL: https://www.spacex.com/launches/starship-flight-12	Official operator event description for partial boostback, engine-light shortfall, landing-burn attempt, and hard splashdown.
DI-1	Uploaded DI Engineering Nonprovisional filing, U.S. Application No. 19/706,930.	Controlled evidence records, candidate buffers, subsystem graphs, audit/replay, authorization matrix, visual anomaly limits, and non-operational release authorization.

3. Source and Evidence Register

Evidence ID	Source	Type	Date / Time	Subsystem Relevance	Support Status	Reliability	Missing Fields	Audit Link
E-001	FAA anomaly statement	Official regulator	May 22, 2026	Booster / flyback	Observed	High	Full telemetry	AUD-F12-E001
E-002	FAA mishap determination	Official regulator	May 27, 2026	Regulatory / safety	Observed	High	Final report, corrective actions	AUD-F12-E002
E-003	FAA public-safety statement	Official regulator	May 27, 2026	Public safety	Observed	High	Debris reconstruction	AUD-F12-E003
E-004	FAA debris response	Official regulator	May 22, 2026	Hazard area / airspace	Observed limited	High	Detailed debris map	AUD-F12-E004
E-005	SpaceX Flight 12 page	Operator update	May 22, 2026	Propulsion / booster	Inferred support	High for event description	Engine telemetry, command logs	AUD-F12-E005
E-006	Public video / webcast	Public visual evidence	Event window	Visual timing	Advisory	Limited	Camera sync, telemetry, geometry	AUD-F12-E006
E-007	Booster telemetry	Not public	Event window	Propulsion / GNC / avionics	Unresolved	Required	All primary telemetry	AUD-F12-E007
E-008	FAA-approved final report	Not found in current public check	Post-event	Root cause / corrective action	Unresolved	Required	Root cause, corrective actions	AUD-F12-E008

The DI framework distinguishes evidence sufficient for observation from evidence sufficient for component-level root-cause commitment, especially where public-source visual evidence lacks telemetry, damage mapping, synchronization, or official investigation data. [DI-1]

4. Event Timeline

Confirmed Timeline

Sequence	Status	Entry
Launch	Observed	Starship Flight 12 launched from Starbase, Texas on May 22, 2026.
Stage separation	Observed	FAA identifies the mishap phase as Super Heavy flyback after stage separation.
Booster anomaly	Observed	FAA states the anomaly involved Super Heavy during flyback over the Gulf of America.
Debris response	Observed limited	FAA activated a Debris Response Area and determined booster debris fell inside the hazard area.
Public-safety result	Observed	FAA reports no public injury or public-property damage.
Mishap determination	Observed	FAA determined on May 27 that the event was a mishap and required a SpaceX-led investigation.

FAA's May 22 and May 27 entries support the confirmed timeline. [FAA-1]

Inferred Timeline

Sequence	Status	Entry
Boostback sequence	Inferred	SpaceX states the booster could not light all planned engines and performed a partial boostback burn that ended early.
Landing-burn sequence	Inferred	SpaceX states Super Heavy attempted to reignite engines for landing burn before hard splashdown.
Propulsion candidate	Inferred / uncommitted	These event descriptions strengthen propulsion burn / engine availability candidates, but do not identify root cause.

SpaceX's official page supports the limited propulsion-sequence description. [SPX-1]

Unresolved Timeline Gaps

Telemetry timeline, engine controller data, thrust and chamber-pressure data, propellant-feed state, GNC response, command-state readiness flags, abort/fallback logic, debris reconstruction, and FAA-approved corrective actions remain unresolved.

5. Hazard Summary

Category	Assessment
Event classification	FAA-determined mishap
Public-safety consequence	No public injury or public-property damage reported by FAA
Vehicle consequence	Super Heavy hard splashdown / booster mishap
Mission-phase risk	Booster flyback / return trajectory
Hazard-area handling	FAA activated Debris Response Area; debris reported inside hazard area
Root-cause status	Unresolved
FAA oversight	FAA-overseen SpaceX mishap investigation required

The event is properly classified as an observed mishap with unresolved root cause. It is not properly classified as a committed propulsion, GNC, avionics, command-state, structural, or ground-support root-cause finding.

6. Observed / Inferred / Unresolved Findings

Category	Findings
Observed	1. FAA determined Flight 12 resulted in a mishap. 2. The mishap involved Super Heavy during flyback after stage separation. 3. No public injury or public-property damage was reported. 4. FAA required a SpaceX-led mishap investigation. 5. FAA will approve the final report and corrective actions. 6. FAA return-to-flight determination depends on public-safety review. [FAA-1]
Inferred	1. Propulsion-side candidate states are strengthened by SpaceX's statement that all planned engines did not light and the boostback burn ended early. 2. Engine relight / engine availability remains a reasonable candidate state, not a committed cause. 3. GNC response may have been affected if propulsion energy state or thrust profile was off-nominal, but telemetry is required. 4. Hard splashdown is a vehicle consequence; cause remains unresolved. [SPX-1]
Unresolved	Root cause, initiating failure mode, engine-health sequence, chamber-pressure behavior, thrust profile, propellant-flow state, command-state alignment, GNC trajectory response, avionics sequence, corrective actions, replay validation, and return-to-flight status remain unresolved.

7. Candidate Causal-State Buffer

ID	Candidate	Status	Supporting Evidence	Missing Evidence
CCS-01	Propulsion burn underperformance	Strengthened / held	FAA mishap + SpaceX partial boostback description	Engine telemetry, thrust, chamber pressure, propellant feed
CCS-02	Engine relight / availability anomaly	Strengthened / held	SpaceX landing-burn attempt before hard splashdown	Engine start commands, relight sequence, controller data
CCS-03	GNC response to off-nominal propulsion	Unresolved	Possible propagation path	Attitude, rates, guidance targets, control output
CCS-04	Command-state / sequencing mismatch	Unresolved	Possible only	Command logs, readiness flags, abort/fallback path
CCS-05	Structural / hard-splashdown consequence	Consequence observed; cause unresolved	FAA + SpaceX hard-splashdown description	Loads, breakup data, debris reconstruction
CCS-06	Ground-support contribution	Unsupported / suppressed	No post-separation support	Ground-system causal evidence

ID	Rule Trigger	Formula Support	Graph Support	Audit	Replay	Commitment	Public Release	Root-Cause Release
CCS-01	Partial	Missing	Present	Placeholder	Not ready	Uncommitted	Limited with caveat	Blocked
CCS-02	Partial	Missing	Present	Placeholder	Not ready	Uncommitted	Limited with caveat	Blocked
CCS-03	Not satisfied	Missing	Present	Placeholder	Not ready	Unresolved	Advisory only	Blocked
CCS-04	Not satisfied	Missing	Present	Placeholder	Not ready	Unresolved	Not as finding	Blocked
CCS-05	Partial	Missing	Present	Placeholder	Not ready	Consequence only	Allowed as consequence	Blocked as cause
CCS-06	Not satisfied	Missing	Weak / none	Placeholder	Not ready	No commitment	Not supported	Blocked

The filing's candidate buffer logic requires candidate causal states to remain uncommitted until input sufficiency, evidence support, rule-trigger support, formula-input support, graph support, audit generation, replay verification, review, and output authorization are satisfied. [DI-1]

8. Subsystem-Interaction Graph

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Propulsion -> GNC [Inferred / unresolved] -> Structures / Aerodynamics [Inferred / unresolved] -> Communications / Telemetry [Observed as evidence dependency] GNC ->
Propulsion command demand [Unresolved] -> Trajectory state [Unresolved] -> Landing / splashdown targeting [Unresolved] Avionics -> Command-State Layer [Unresolved] ->
Propulsion sequencing [Unresolved] -> Telemetry [Unresolved] Command-State Layer -> Readiness flags [Unresolved] -> Engine-start authorization [Unresolved] -> Abort /
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fallback / override paths [Unresolved] Structures / Aerodynamics -> Vehicle load state [Unresolved] -> Hard splashdown consequence [Observed consequence] -> Debris behavior [Observed limited] Communications / Telemetry -> Evidence sufficiency [Observed gap] -> Replay validation [Unresolved] -> Root-cause auditability [Unresolved] Ground Support -> Flyback anomaly [Unsupported / blocked] Regulatory / Public-Safety Layer -> FAA mishap status [Observed] -> Corrective-action gate [Observed requirement / unresolved substance] -> Return-to-flight gate [Blocked pending official determination]
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9. Propagation Records

ID	Source	Destination	Mechanism	Evidence Support	Diagnostic Status	Commitment
PR-01	Propulsion	Return-energy state	Partial boostback / burn underperformance	SpaceX event description	Inferred	Blocked from root-cause commitment
PR-02	Propulsion	GNC	Off-nominal thrust profile increasing control demand	No telemetry	Unresolved	Blocked
PR-03	GNC	Structures / aerodynamics	Trajectory or attitude divergence leading to splashdown loads	No GNC data	Unresolved	Blocked
PR-04	Avionics / command-state	Propulsion	Sequencing or readiness-state impact on relight	No command logs	Unresolved	Blocked
PR-05	Hard splashdown	Debris / hazard-area response	Vehicle consequence and debris response	FAA statement	Observed consequence	Committed only as consequence
PR-06	Ground support	Flyback anomaly	Ground-system contribution after stage separation	No support	Unsupported	No commitment

10. Engineering Integrity Metrics

Metric	Rating	What It Supports	What It Cannot Support
Evidence sufficiency	3 / 5 event; 1 / 5 root cause	FAA mishap status and public-safety outcome	Component-level cause
Telemetry integrity	0 / 5 public	Identifies missing replay inputs	Any causal commitment
Command-state alignment	0 / 5 public	Identifies required data gap	Command fault finding
Propulsion divergence	2 / 5	Candidate propulsion review	Root cause without telemetry
GNC trajectory divergence	1 / 5	Candidate propagation path	GNC fault finding
Subsystem propagation load	3 / 5 candidate	Multiple plausible paths	Final causal chain
Public-source visual reliability	1 / 5 cause; 3 / 5 visible consequence	Timing / visible consequence	Component cause
Replay readiness	1 / 5	Packet skeleton	Deterministic verification
Audit completeness	2 / 5	Evidence IDs and candidate IDs	Full replay audit
Release authorization	Advisory-only	Public factual release	Root-cause or return-to-flight release

The framework supports diagnostic states such as observed, inferred, unresolved, engineering review required, export held, release authorized, and release blocked, and it maps authorization states through an authorization matrix. [DI-1]

11. Replay-Ready Diagnostic Packet

Replay Field	Entry
Packet ID	F12-SH-RDP-v1.0
Input records	E-001 through E-008
Rule configuration	F12-LV-RULESET-placeholder-v0.1
Formula configuration	PROP-DIV, GNC-DIV, CMD-ALIGN, TELEMETRY-SUFF, REPLAY-READY
Threshold table	Missing: thrust, chamber pressure, relight timing, attitude/rate limits

Replay Field	Entry
Subsystem graph version	F12-SH-FLYBACK-GRAPH-v0.2
Candidate IDs	CCS-01 through CCS-06
Propagation IDs	PR-01 through PR-06
Audit fields	Evidence IDs, candidate IDs, rule IDs, formula IDs, graph edges, diagnostic state, authorization state
Replay validation	Not ready
Verification hash	HASH-PENDING-F12-SH-REPLAY-001

Structured audit records and replay specifications in the filing are designed to link input records, telemetry identifiers, rule identifiers, formula identifiers, graph identifiers, candidate states, propagation records, diagnostic states, replay values, and release authorization results. [DI-1]

12. Replay Delta Review

Baseline Claim	New / Current Evidence	Source	Change Type	Authorization Effect
FAA confirmed mishap	FAA statement remains controlling	FAA	No change	Observed fact release-authorized
No public injury / property damage	FAA statement remains controlling	FAA	No change	Public-safety fact release-authorized
FAA investigation required	FAA May 27 requirement remains controlling	FAA	No change	Root-cause release blocked
Return to flight depends on FAA public-safety determination	FAA states return-to-flight basis	FAA	No change	Return-to-flight conclusion blocked
Propulsion candidate	SpaceX official page supports partial boostback and hard splashdown sequence	SpaceX	Strengthened	Candidate remains uncommitted
Final report unavailable	No FAA final Flight 12 closure located in current official check	FAA public status review	No change	Investigation pending

The meaningful delta from v0.1 to v1.0 is that the propulsion-side candidate remains strengthened by SpaceX's event description, but it still cannot be committed as root cause without telemetry and FAA-approved findings. [SPX-1]

13. Post-Mishap Corrective-Action Gate

Candidate	Required Corrective-Action Evidence	Verification Evidence	Test Evidence	Closure Status	Release Effect
CCS-01 Propulsion burn underperformance	Engine-system corrective action, affected engine set, burn-profile fix	Engine telemetry, thrust, chamber pressure, propellant feed	Static-fire, relight validation	Open	Advisory only
CCS-02 Engine relight / availability	Relight-sequence correction, controller review	Engine-start commands, ignition margins	Relight test data	Open	Advisory only
CCS-03 GNC response	GNC update or no-fault finding	Attitude, rates, trajectory solution, control authority	GNC simulation / flight replay	Open	Export held
CCS-04 Command-state sequencing	Command-state audit, readiness validation	Command logs, readiness flags, override path	Replay of sequence logic	Open	Root-cause blocked
CCS-05 Hard splashdown consequence	Splashdown analysis, debris reconstruction	Loads, breakup sequence, debris map	Structural / hydrodynamic review	Partially supported as consequence	Public factual release with caveat
CCS-06 Ground support	Only if official evidence links ground support	Pad / ground-system causal evidence	Ground-system test records	Suppressed	No release as finding

14. Return-to-Flight Authorization Matrix

Condition	Public Fact Release	Advisory Output	Engineering Review	Export Held	Root-Cause Blocked	RTF Blocked	Conditionally Reviewable
FAA mishap determination	Yes	Yes	Optional	No	N/A	Yes	No

Condition	Public Fact Release	Advisory Output	Engineering Review	Export Held	Root-Cause Blocked	RTF Blocked	Conditionally Reviewable
No public injury/property damage	Yes	Yes	Optional	No	N/A	Does not clear RTF	No
SpaceX partial boostback description	Yes, limited	Yes	Required	For causal finding	Yes	Yes	Yes
Propulsion causal candidate	Limited	Yes	Required	Yes	Yes	Yes	Yes
GNC candidate	No as finding	Yes	Required	Yes	Yes	Yes	Yes
Command-state candidate	No as finding	Yes	Required	Yes	Yes	Yes	Yes
Ground-support cause	No	No	Required if evidence appears	Yes	Yes	Yes	No
FAA-approved final report	If available	Yes	Required	No if accepted	May unlock	May unlock	Yes
FAA return-to-flight determination	If available	Yes	Required	No if accepted	Depends on scope	May unlock	Yes

Release authorization in the filing controls diagnostic-output release only and does not authorize ignition, launch, flight, test execution, operational command execution, regulatory approval, or safety certification. [DI-1]

15. Residual Risk Register

Risk	Current State	Residual Exposure	Missing Evidence	Closure Condition
Propulsion recurrence risk	Inferred / uncommitted	Open	Engine telemetry, corrective actions	FAA-accepted corrective action + verification
GNC response risk	Unresolved	Open	Attitude, rates, trajectory, control data	GNC replay / official no-fault finding
Command-state recurrence risk	Unresolved	Open	Command logs, readiness flags, override path	Command-state audit closure
Telemetry sufficiency risk	High	Open	Synchronized telemetry package	Replay-ready evidence package
Public-safety risk	FAA reports no public injury/property damage; gate remains active	Review required	FAA final public-safety determination	FAA return-to-flight determination
Replay/audit risk	High	Open	Replay specification, hash, full audit links	Deterministic replay validation
Public-source evidence confusion	Moderate	Overclaiming risk	Camera sync, telemetry, official findings	Strict observed/inferred/unresolved separation

16. Final Authorization State

Category	State
Cleared for public factual release	FAA mishap determination, Super Heavy flyback phase, no public injury or public-property damage reported, FAA Debris Response Area activation and hazard-area outcome, SpaceX limited event description of incomplete boostback / landing-burn attempt, and FAA investigation requirement.
Advisory only	Propulsion burn underperformance candidate, engine relight / availability candidate, possible GNC propagation, possible command-state review need, and structural hard-splashdown consequence analysis.
Blocked	Root cause, corrective-action sufficiency, committed propulsion fault, committed GNC fault, committed command-state fault, ground-support cause, replay-verified closure, and return-to-flight clearance.
Investigation-pending	FAA-overseen SpaceX mishap investigation, final report, corrective actions, and any return-to-flight public-safety determination.
Closure-ready evidence	FAA-approved final mishap report, FAA-accepted corrective actions, SpaceX corrective-action package, synchronized engine / GNC / command telemetry, debris reconstruction, replay specification, audit record, and deterministic verification value.

17. Appendix A - Evidence Gate Checklist

Gate	Status	Effect
FAA final mishap report	Not found in current public check	Blocks root-cause closure
FAA corrective-action approval	Not found in current public check	Blocks corrective-action sufficiency

Gate	Status	Effect
SpaceX corrective-action summary	Not found in current official check	Blocks closure-ready status
Engine telemetry	Missing	Blocks CCS-01 / CCS-02
GNC telemetry	Missing	Blocks CCS-03
Command-state logs	Missing	Blocks CCS-04
Replay validation package	Missing	Blocks replay closure
Debris / splashdown reconstruction	Missing	Limits CCS-05 to consequence
Updated license / return-to-flight determination	Not established from current public check	Blocks return-to-flight conclusion

18. Appendix B - Audit Trail Skeleton

Record Type	Placeholder IDs
Evidence records	AUD-F12-E001 through AUD-F12-E008
Candidate causal states	AUD-F12-CCS-001 through AUD-F12-CCS-006
Propagation records	AUD-F12-PR-001 through AUD-F12-PR-006
Rule-trigger records	RULE-F12-RT-001 through RULE-F12-RT-006
Formula records	FORM-F12-PROP, FORM-F12-GNC, FORM-F12-CMD, FORM-F12-TEL
Replay records	REPLAY-F12-v1.0 / HASH-PENDING-F12-SH-REPLAY-001
Authorization records	AUTH-F12-PUBLIC, AUTH-F12-ADVISORY, AUTH-F12-ROOT-BLOCK, AUTH-F12-RTF-BLOCK

19. Appendix C - Non-Commitment Language

"This report does not determine root cause."

"Public visual evidence supports timing and visible consequence only."

"Candidate causal states remain uncommitted absent telemetry, investigation findings, and authorization support."

"Return-to-flight conclusions remain blocked unless official regulatory evidence supports them."

"Release authorization in this report applies only to diagnostic-output handling and does not authorize launch, flight, test execution, regulatory approval, or safety certification."

20. Appendix D - Deterministic Visual Scan on Rocket Image

The uploaded rocket image is treated as visual context only and is not used as root-cause evidence.

Figure D-1. Uploaded launch image used for deterministic visual scan. Image-only review; not a root-cause source.

Field	Deterministic Scan Result
Visible object	Large launch vehicle ascending near launch tower
Mission phase visible	Liftoff / early ascent
Visible plume	Large exhaust plume and ground-cloud formation
Launch infrastructure	Tower / pad structures visible
Apparent vehicle state	Vehicle appears upright and ascending
Visible anomaly	No definitive anomaly identifiable from still image alone
Explosion / breakup evidence	Not confirmed from image alone
Root-cause evidence	None

Class	Image-Only Assessment
Observed	Rocket is in early ascent or liftoff phase; major exhaust plume and pad-adjacent cloud formation are visible; launch tower and support structures are visible; vehicle remains visibly intact in the frame.
Inferred, low confidence	Image is consistent with a heavy-lift launch or test flight; plume/cloud morphology is consistent with high-thrust engine operation near the pad.
Unresolved	Exact launch event, date/time, configuration, engine health, off-nominal condition, and causal relationship to Flight 12 flyback anomaly.

Subsystem	Image Evidence	Diagnostic State
Propulsion	Exhaust plume visible	Observed operation; no fault committed
GNC	Vehicle appears upright	Inferred stable attitude; insufficient for GNC finding
Structures	Vehicle appears intact	Observed visible integrity only
Avionics	Not visible	Unresolved
Command-state layer	Not visible	Unresolved
Ground support	Tower and pad structures visible	Observed context only
Telemetry	Not present	Missing
Public safety / debris	No visible debris field confirmed	Unresolved

Candidate	Image Effect
CCS-01 Propulsion burn underperformance	Not assessable from this still image
CCS-02 Engine relight / availability anomaly	Not assessable; image appears liftoff-phase, not flyback relight
CCS-03 GNC response	Not assessable beyond apparent upright ascent
CCS-04 Command-state sequencing	Not visible
CCS-05 Hard-splashdown / structural consequence	Not depicted
CCS-06 Ground-support contribution	No causal support from image alone

Appendix authorization result: Image scan status: advisory only. Observed visual facts: release authorized. Anomaly finding: blocked. Root-cause finding: blocked. Flight 12 causal linkage: blocked. Telemetry requirement: active. Replay-ready status: not ready.

Non-commitment note: Under the DI Engineering framework, public-source visual evidence can support geometry, timing, and visible-state observations while preserving unresolved classification for root cause absent telemetry, official investigation data, audit linkage, and replay validation. [DI-1]

Final Determination

Report status: Valid as an audit-ready advisory diagnostic report. Root cause unresolved. Corrective-action sufficiency not determined. Return-to-flight not determined from public record. Public factual release allowed for verified FAA/SpaceX facts; causal findings remain advisory-only and export-held.